A 45-Year Global Analysis of the Spatial Human-Forest Nexus

Response to reviewers

Emanuele Massaro, Peter Newton, Juan Carlos Ciscar, Gregoire Dubois, Angela Fanelli, Dolores Ibarreta, Nicola Riccetti, Wojciech Szewczyk and Alessandro Cescatti

Dear Reviewers,

We extend our sincere gratitude to both reviewers for their insightful critiques, which have greatly strengthened the manuscript and the overall impact of this research. We have carefully revised the manuscript to incorporate all feedback provided. Below, you will find a detailed point-by-point response outlining the changes made to the text. The reviewers' original comments are included in blue text, and we have endeavored to address each one thoroughly, clarifying the rationale behind our methodology and the validity of our findings.

Key revisions include:

1. Enhanced analysis of global interactions between forests and human systems, with clarified explanations of the indices used and expanded discussions of their spatiotemporal dynamics, contextualized within relevant literature and supplementary data.
2. Revised title to better reflect the spatial dimension of the proposed Forest-Human Nexus metric.
3. Strengthened robustness of the Forest Proximate People analysis, accounting for its sensitivity to varying distance thresholds (as highlighted by Reviewer #1).
4. Critical contextualization of findings in relation to current literature and case studies (as requested by Reviewer #2).

Thank you once again for your invaluable contributions to improving this work.

Sincerely,

Emanuele Massaro, Peter Newton, Juan Carlos Ciscar, Gregoire Dubois, Angela Fanelli, Dolores Ibarreta, Nicola Riccetti, Wojciech Szewczyk, and Alessandro Cescatti

Reviewer #1 (Remarks to the Author):

Dear Editor and authors.

I was pleased to review the manuscript ‘Spatiotemporal Patterns in the Human-Forest Nexus: A 45-Year Global Analysis’ and I would like to express my appreciation to the authors for their renewed focus on ‘forest-human’ spatial proximity since their article ‘The Number and Spatial Distribution of Forest Proximate People Globally’. I appreciate the author's renewed focus on “forest-human” spatial proximity after old work. I believe this manuscript has potential for publication in Communication Earth & Environment, even though there is still much work to be done to complement and improve it.

We thank Reviewer 1 for appreciating our work from a scientific viewpoint and for their relevant comments that helped us to significantly improve the manuscript.

1. In contrast to the author's previous publication ‘The Number and Spatial Distribution of ForestProximate People Globally’, this manuscript uses only ‘Forest-proximate people’. The manuscript only uses a single definition of ‘Forest proximate people’, and it is difficult for me to know whether FPP still stands for ‘The Number and Spatial Distribution of Forest Proximate People Globally’ as defined in the article ‘The Number and Spatial Distribution of Forest Proximate People’. I have difficulty knowing whether FPP still stands for consistency with the definition in ‘The Number and Spatial Distribution of Forest Proximate People Globally. In addition, I suggest that the authors conduct sensitivity analyses of FPP at different distances, such as 1km, 2km, 3km, 4km, 5km,and other distance to analyse the impact of the different definitions of Forest Proximate People on the results and to determine whether the choice of other distances would change the robustness of the results. The sensitivity analysis of FPP at different distances was carried out to analyse the effects of different definitions of ‘Forest Proximate People’ on the results of the study, to capture the human-forest relationship more comprehensively.

We thank the Reviewer for raising this important point, which has allowed us to clarify the rationale behind our methodological choices. We acknowledge that the selection of a distance threshold—as highlighted in prior literature (Newton, P., et al. "The number of forest-and tree-proximate people” (FAO 2022), Newton, Peter, et al. "The number and spatial distribution of forest-proximate people globally." One Earth 3.3 (2020): 363-370.)—can indeed be arbitrary, often tailored to specific regional or contextual needs. Consistent with this understanding, our framework intentionally avoids prescribing a universal threshold. Instead, we emphasize the adaptability of our open-source methodology (as shared in the public repository linked to this paper), enabling users to define distances (e.g., 1 km, 5 km, etc.) that best align with their objectives. Please see lines 152 and 162 of the revised manuscript where we better explained this point, as well as the analysis below and the Section S6 “*FPP distance analysis”* in the SI of the revised submission.

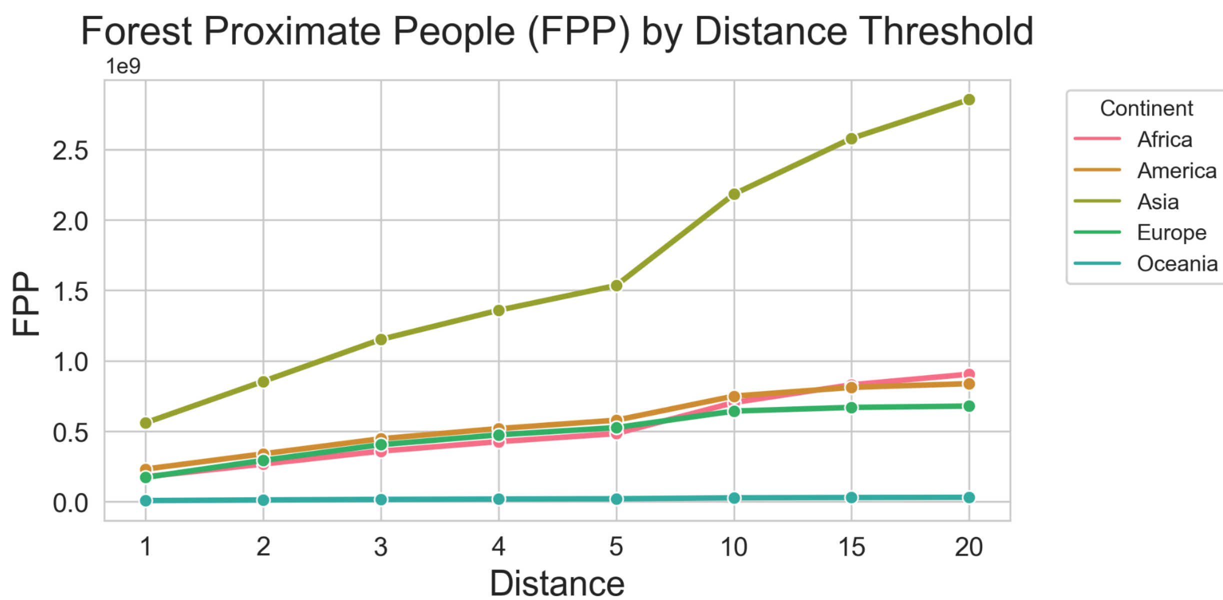

Figure 1. FPP by distance for 2020 for different continents.

As suggested, we conducted sensitivity analyses for FPP across multiple distance thresholds (d_t = 1, 2, 3, 4, 5, 10, 15, 20 km). Our goal was to determine whether the relative trends in FPP over time (quantified by Sen's Slope) depend on the choice of d_t. While, if we increase the distance, it is normal that the FPP increase as shown for 2020 in Figure1, we are interested to see if the temporal variation of the trend of the FPP changes significatively.

Key Results:

- Rank Stability (Spearman Correlation):
    - Pairwise correlations between rankings across d_t groups are exceptionally high (e.g., ρ = 0.96 between d_t=1 and d_t=2, rising to ρ = 1.0 for larger thresholds like d_t=10–20).

Spearman Rank Correlations:

d_t	1	2	3	4	5	10	15	20
d_t								
1	1.00	0.96	0.91	0.84	0.80	0.70	0.70	0.70
2	0.96	1.00	0.98	0.91	0.89	0.80	0.79	0.79
3	0.91	0.98	1.00	0.97	0.96	0.88	0.86	0.86
4	0.84	0.91	0.97	1.00	0.99	0.92	0.90	0.90
5	0.80	0.89	0.96	0.99	1.00	0.95	0.93	0.93
10	0.70	0.80	0.88	0.92	0.95	1.00	1.00	1.00
15	0.70	0.79	0.86	0.90	0.93	1.00	1.00	1.00
20	0.70	0.79	0.86	0.90	0.93	1.00	1.00	1.00

    - This indicates near-identical rankings of regions by FPP trends across all tested thresholds (see heatmap below in Figure 2).

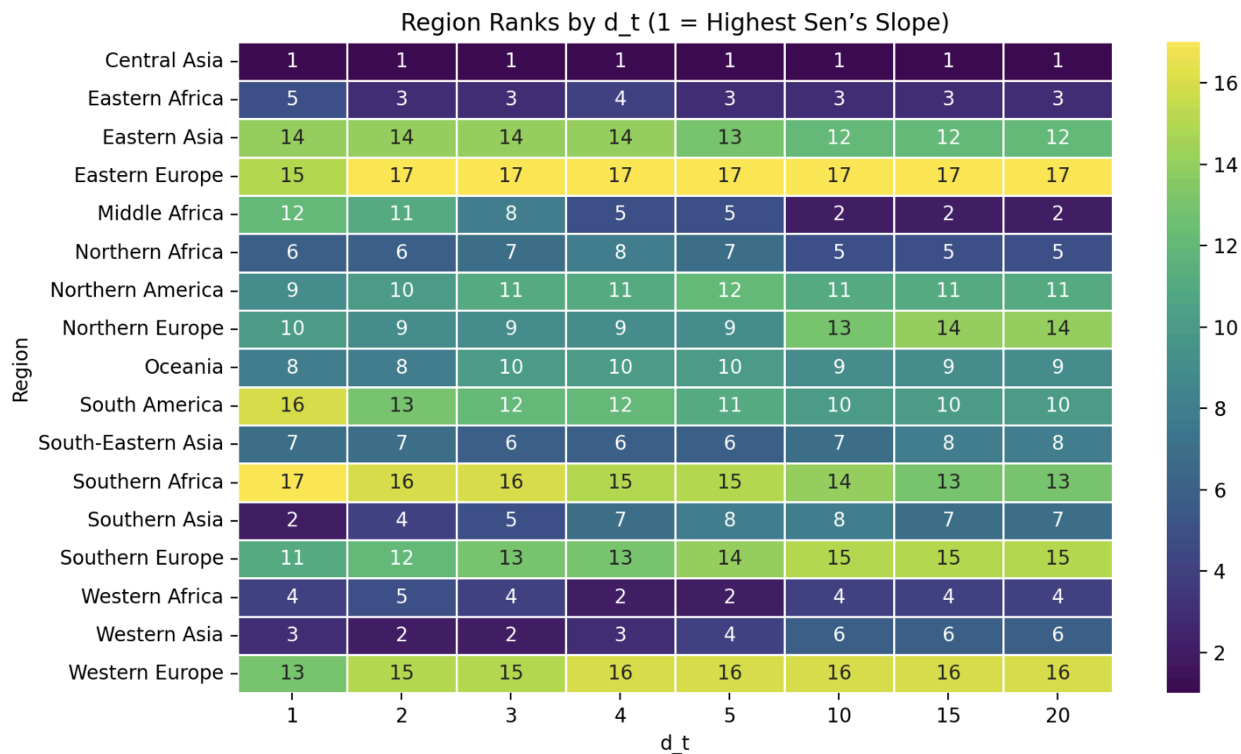

Figure 2. Plot ranks across d_t groups.

- Kendall's Concordance (W = 0.901):
  - Strong consensus in rankings across all d_t groups ($W \approx 0.9$, where 1 = perfect agreement).
  - Example: Central Asia consistently ranks 1st (strongest positive trend), while Southern Africa and Western Europe remain at the bottom across all thresholds.

- Friedman Test ($p = 0.99$):
  - No statistically significant differences in rankings across d_t groups ($p > 0.05$), confirming robustness.

Our analysis of Forest-Proximate People (FPP) trends across multiple distance thresholds (1–20 km) reveals a striking consistency in the relative trajectories of human-forest proximity globally. Despite varying the spatial definition of "forest proximity" (as captured by d_t), the rankings of regions by their Sen's slope values—a measure of FPP trend strength—remain nearly invariant. For instance, Central Asia consistently exhibits the strongest positive trend across all thresholds, while Southern Africa and Western Europe persistently rank lowest. This invariance is statistically robust:

- Kendall's Concordance Coefficient (W = 0.901) demonstrates near-perfect agreement in regional rankings across thresholds, indicating that the choice of d_t does not meaningfully alter conclusions about which regions are experiencing the most rapid changes in FPP.

- High Spearman correlations ($\rho > 0.95$ for adjacent thresholds) further confirm that even small adjustments to the proximity definition (e.g., 1 km vs. 5 km) yield virtually identical regional prioritizations.
- The Friedman test ($p = 0.99$) conclusively rejects the hypothesis that rankings differ significantly across thresholds.

These results affirm that the human-forest relationship, as quantified by FPP trends, is fundamentally robust to the spatial threshold used to define "proximity" within the tested range (1–20 km). While absolute FPP counts vary with d_t—as expected when expanding or contracting buffer zones—the relative trajectories of regions remain stable. This suggests that underlying drivers of FPP dynamics (e.g., deforestation fronts, urbanization patterns, or policy interventions) operate at scales larger than the tested thresholds, and their impacts are detectable regardless of the precise boundary definition.

We also computed an elasticity measure to show how sensitive FPP (Forest Proximate People) is to changes in distance. It quantifies how much FPP changes when we slightly increase the distance threshold and whether there is a strong or weak relationship between the number of people near forests and the chosen distance. A high elasticity means that increasing the distance threshold significantly increases FPP, whereas a low elasticity means that increasing the distance threshold has little effect on FPP. To compute elasticity, we calculated the percentage change in FPP as the distance threshold increased for each region and year. This provided insight into how quickly the population count rises as we expand the proximity threshold. We also computed the step-wise increase in distance to understand the incremental effect on FPP. Elasticity was then computed as the ratio of the percentage change in FPP to the change in distance:

$$Elasticity = \frac{\%\Delta FPP}{\Delta d_t}$$

This formulation allows us to understand how much FPP changes per kilometer increase in distance.

The elasticity analysis provided (Figure 3) key insights into the behavior of FPP at different distances. At smaller distances (e.g., 1-5 km), elasticity is typically high, meaning that small increases in distance lead to large increases in FPP. However, at larger distances (beyond 10 km), elasticity declines, indicating that increasing the distance further adds fewer additional people. This suggests that the biggest impact of distance on FPP happens within the first few kilometers, supporting 5 km as a meaningful threshold.

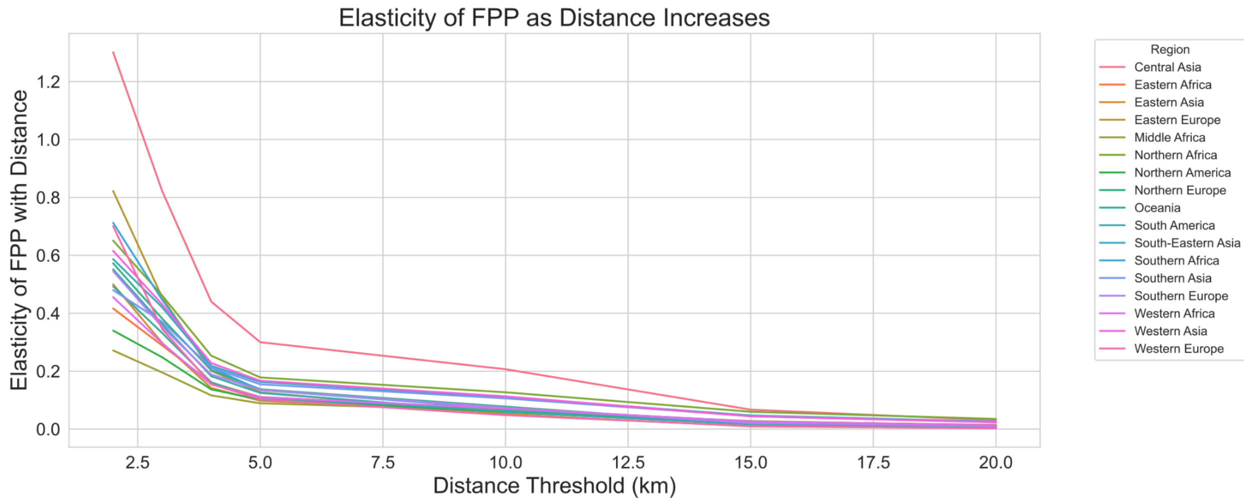

Figure 3. Average values of elasticity across regions and distances between 1975 and 2020.

This analysis could justify our choice of 5 km as a reasonable cutoff.

2. The authors have developed a new concept of 'Forest Human Nexus', but I believe that the Forest-Human relationship should include the contribution of forests to human livelihoods or ecosystem services. For example, forests can provide wood energy, forest food, and recreational services; and human activities can interfere with forests (forest managers may improve the forest ecosystem, or deforestation may reduce forest area and quality). Therefore, I believe that the current 'Forest Human Nexus' is not yet rigorous as it only includes forest area per capita, forest accessibility, and population density. Therefore, I suggest that the authors clarify the term 'Nexus', such as the level of spatial proximity, i.e., the definition should be geospatial rather than aggregate.

We appreciate the reviewer's insightful comments regarding the conceptualization of the 'Forest Human Nexus.' We recognize the importance of considering the functional relationships between forests and humans, including ecosystem services like wood energy, food provision, and recreational opportunities, as well as human impacts such as forest management and deforestation.

Our study focuses on the **spatial relationship** between forests and human populations over time, rather than a functional or ecosystem services-based framework. Specifically, we analyze **forest area per capita, forest accessibility, and population density** to capture **spatial patterns** in the human-forest connection. We acknowledge that functional interactions are an important aspect of forest-human relationships, but they are beyond the scope of this study. The myriad different contributions of forests to people's livelihoods is captured in Newton et al. (2016) *Land Use Policy*. Spatial proximity is one commonality that defines many (through not all) human-forest relationships, and so we focus on the spatial relationship as being an often-necessary but rarely-sufficient descriptor of human-forest interactions.

To enhance clarity, we have revised the manuscript throughout (from the Title to the Abstract, and from the Introduction to the Discussion) to explicitly state that the ‘Forest Human Nexus’ in our study is **a spatial concept**, rather than an aggregate or functional assessment. Additionally, we have updated the title to *"A 45-Year Global Analysis of the Spatial Human-Forest Nexus"* to more accurately reflect our focus.

3. In the title of the article, the author uses the term ‘45 years’, I think that the study concentrates too much on the years 1975 and 2020, and even though it is covered in the supplementary material, less attention is paid to the evolution of the forest-human relationship in other periods in the body of the text, and I would suggest that the author add the data of the important turning years to highlight the fact that the long time series of ‘45 years’ is a long period. I suggest that the authors add data on important turning years to highlight the special features of the ‘45 years’ long time series study.

Thank you for your valuable comment regarding the emphasis on the full 45-year timeframe in our study. We appreciate your suggestion to highlight the evolution throughout the entire period rather than focusing primarily on the endpoints (1975 and 2020). In our revised version, we have specifically addressed this concern by including a comprehensive temporal analysis of all three variables across the entire 1975-2020 period, with particular attention to this aspect in the Supplementary Information (SI). In the SI, we present detailed analyses of temporal patterns, including growth and decline trends, which we believe provide more valuable insights than isolated turning points.

While some regions do show significant shifts or important changes at specific years, we have focused on capturing the continuous evolution of forest-human relationships throughout the entire 45-year timeframe. This approach allows us to better understand the long-term dynamics rather than emphasizing only specific years.

Based on your feedback, we have added clearer references in the main text to direct readers to these detailed temporal analyses in the Supplementary Information (see subsections Temporal Analysis in SI for each metric), ensuring that the value of our long-term study is fully appreciated.

In particular, we analyzed the relative temporal variation from 1975. In Figure 4 (and in Section S4 in SI of the revised manuscript) of this response we report the example of temporal variation of FHN defined as the relative change of the median values of FHN across regions. To identify significant shifts in these temporal patterns, we determined turning points (highlighted as red dots in Figure 4) using a methodology that detects points where the slope changes sign and exceeds a specific threshold. Regarding the methodology used to identify turning points, we adopted an approach based on analyzing the slopes between consecutive points in the time series. In simple terms, we calculate the rate of change (slope) between each pair of consecutive years. A turning point is identified when the direction of the slope changes (from positive to negative or vice versa) and the difference between the slopes exceeds a predefined minimum threshold. This method allows us to identify moments when there was a significant change in the trend direction, while filtering out small fluctuations that might represent noise in the data rather than true changes in the forest-human dynamics

These turning points represent significant changes in the trajectory of forest-human relationships over the 45-year period, helping to identify key moments of transition in different regions. This approach allows us to not only observe the overall trends but also pinpoint critical periods where the nature of forest-human dynamics underwent substantive changes.

We now discuss this in Section S4 of the revised version of the manuscript.

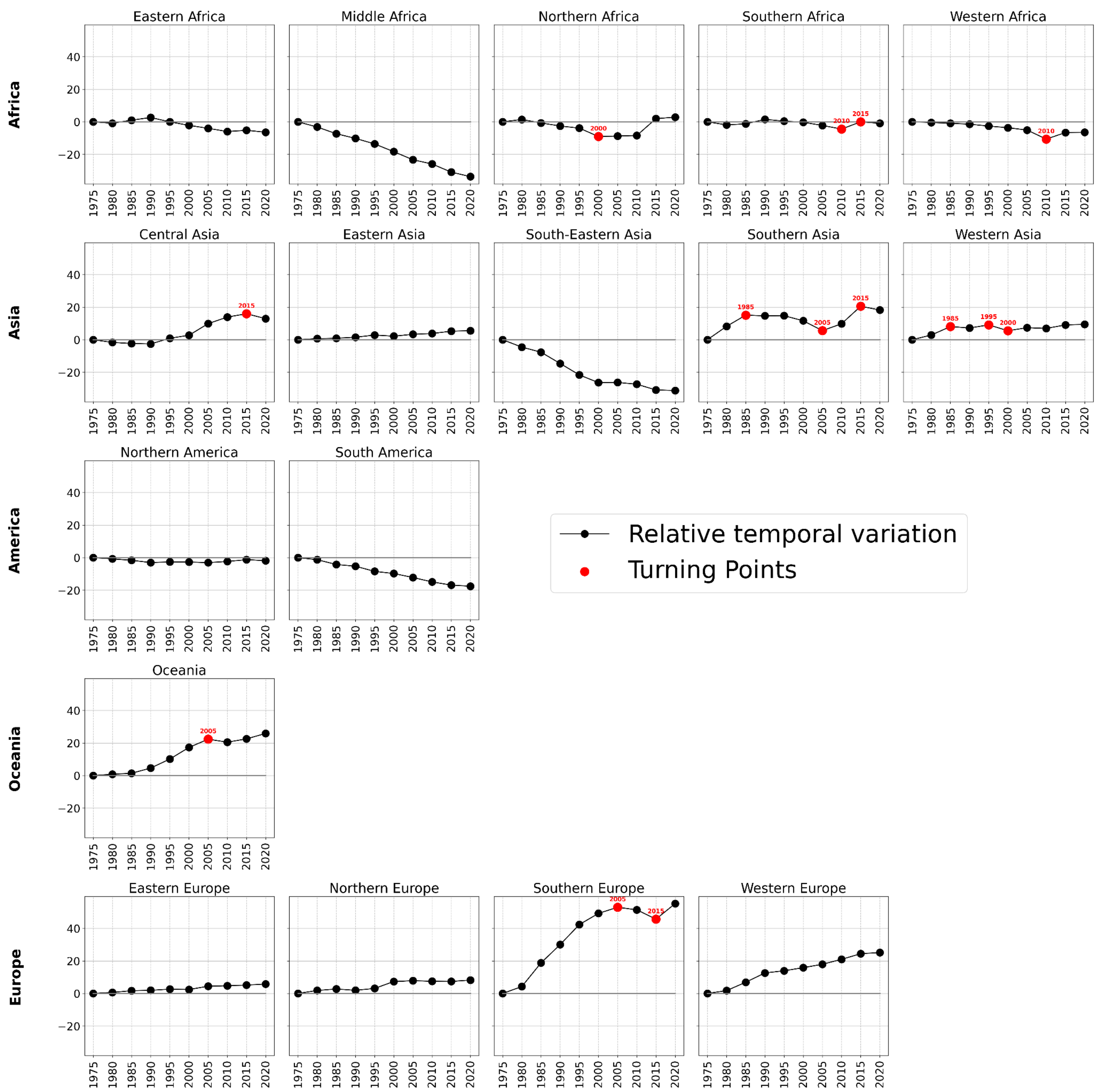

Figure 4. Temporal variation of FHN

4. In the introduction, the authors' literature review focuses on forest ecosystem services, which tends to focus on small-scale studies. I believe that the core of the authors' paper is the “Forest-Human Nexus” aspect, and therefore I suggest adding spatial Human-Forest interactions, such as changes in human communities close to forests, changes in “FPP”, etc.

We thank the reviewer for this valuable suggestion to enhance our introduction with more focus on spatial Human-Forest interactions. Following this recommendation, we have added a substantial new paragraph to our introduction (see lines 34-51 of the revised manuscript) that specifically addresses the dynamic spatial relationships between humans and forests.

The new text discusses how human-forest spatial configurations have evolved in different regions due to land use changes, urbanization, and demographic shifts. We have included specific examples from the Brazilian Amazon and Southeast Asia to illustrate these changing spatial relationships. We have also expanded on the Forest Proximate People (FPP) concept, noting regional variations and temporal changes in forest-adjacent populations.

This addition strengthens the introduction by:

1. Directly addressing the "Forest-Human Nexus" spatial dimension that is central to our paper
2. Providing context for understanding changes in FPP patterns globally
3. Highlighting the importance of spatial analysis in understanding human-forest relationships
4. Supporting these points with seven additional references focused specifically on spatial aspects of human-forest interactions

This enhancement creates a stronger foundation for our analysis of the Forest Human Nexus as a spatial indicator and better positions our work within the existing literature on spatial human-forest relationships.

5. In Line 71-76, I think that the authors have written a very vague contribution to the study. Specifically, it is important to understand whether the Forest-Human Nexus will contribute to changes in ‘community forest’ policies, support improvements in human well-being in areas adjacent to forests, contribute to the timely achievement of Sustainable Development Goals, and contribute to the development of a sustainable forest. I hope that the authors will be able to combine these aspects in their contribution to the study.

We appreciate the reviewer's suggestion to clarify the contribution of our study. We wish to emphasize that the primary objective of this research is to develop and validate novel metrics for quantifying human-forest spatial relationships over time, particularly the Forest-Human Nexus (FHN) index. These metrics provide foundational analytical tools that can support various applications rather than directly prescribing specific policy interventions.

The FHN and other metrics introduced in this study offer standardized, globally consistent measurements that can serve multiple purposes. They can be used to: (1) evaluate the effectiveness of community forest policies by tracking changes in human-forest proximity over time; (2) identify regions where declining forest accessibility may impact human well-being, thus requiring targeted interventions; (3) monitor progress toward specific SDGs, particularly those related to forest conservation (SDG 15) and sustainable communities (SDG 11); and (4) inform sustainable forest management by revealing changing patterns in human-forest relationships.

While developing specific policy recommendations is beyond the scope of this study, we have added text in the Discussion section (lines 340-350 of the revised manuscript) highlighting how these metrics can be applied by policymakers, conservation practitioners, and researchers to address the important concerns raised by the reviewer. Our contribution lies in providing robust analytical tools that enable more informed decision-making across these domains.

6. In the results section, the authors briefly describe their main findings, however, I found that many of them are qualitative and lack some good quantitative descriptions. For example, in Line 79, it is difficult to get a clear picture of the changes in 'FAP' and the key inflexion points, which leads to the absence of some key information of interest. I suggest the authors to add the key information.

We thank the Reviewer for this comment: by following comment 3 we added information of the temporal patterns of all the variables, the SI of the revised manuscript and comments in the main text (we changed all the Sections in the Results to address this point, "Global analysis of Forest Area per Person", "Temporal patterns of Forest Proximate People" and "Temporal patterns of Forest Human Nexus".

7. Many of the author's graphs have only the starting and ending years 1975 and 2020, while some of the graphs for key years are in the supplementary material. I suggest that key years be added to the text. In addition, the clarity (DPI) of the figures needs to be further increased; the illustrations in the manuscript are not clear. I think the authors need longer figure titles, and in its current state, some of the detailed information in the figure is not clearly illustrated and is not consistent with the Communication Earth & Environment publishing style.

We apologize: in the manuscript we included only low-resolution images, and provided the high resolution images as separate files. In the revised version we have included high-resolution images in the main manuscript.

8. In Line 128-129, I suggest not putting equations in the results section, but in the methods section.

We thank the Reviewer for this comment. We changed it as suggested.

9. The author's analyses of 'FAP' and 'FPP' mostly focus on the analysis of spatial and temporal patterns of change. However, it is difficult to reveal the trend of change in the time series by a single analysis of spatial and temporal patterns. I suggest that the authors can use the method of trend analysis to analyse the change patterns of FAP and FPP in the global image metric scale. As far as I know, such as the BFAST algorithm, slope, MK-SEN, and key inflexion points can be further identified to strengthen the understanding of the 'Forest-Human' relationship.

We thank the Reviewer for this point. As suggested in Comments 3 and 6 we have added the temporal analysis part in the SI of the revised manuscript and comments in the main text.

10. Even though I agree with the authors' important effort to define "FPP" in "The Number and Spatial Distribution of Forest Proximate People Globally", I am not convinced that this is the right definition for "FPP". Even though I agree with the author's important effort to define 'FPP' in 'The Number and Spatial Distribution of Forest Proximate People Globally', I would like to request that a detailed definition of the term 'FPP' be provided in the manuscript to strengthen the validity of the concept of 'FPP'.

We defined Forest Proximity Population at distance d (FPPd) as the number of people living within a given distance (d) to a forest. Using a global moving window approach with 50kmx50km grid cells, we identified forest pixels within each window. For windows containing at least one forest pixel, we calculated the total population living within distance d from these forest pixels. This methodology enables quantification of forest-adjacent human populations at various distances, providing a spatially explicit measurement of population distribution relative to forest resources worldwide (see Methods section).

11. In Figure 4, the authors focus on the regions of Southeastern Asia and Southern Asia. However, in the text, some other regions are mentioned, and I think that the Amazon region and the Congo Basin region should be shown in the text, and Figure 4. C, it is suggested to divide the regions according to national scales, as the current scales are different, and it is necessary to be consistent.

We thank the reviewer for this valuable suggestion. We would like to clarify that all metrics in our analysis were computed using a consistent 50km × 50km resolution across all regions globally. This uniform spatial resolution forms the foundation for all maps presented in Figures 1-6 of the new revised manuscript. Regional aggregations were performed only after this consistent spatial analysis to illustrate broader trends.

Upon further consideration and in response to reviewer feedback, we have decided to move the old Figure 4 (now Figure S19) from the main manuscript to the SI because it was not adding relevant information to the other analysis. We determined that the shift analysis presented in this figure largely duplicated information already captured in the relative change analysis shown in the Supplementary Information. To streamline the presentation of our findings, we have incorporated the relevant insights from the shift analysis into the Supplementary Information section, ensuring that no substantive content is lost while improving the overall clarity and focus of the manuscript.

12. I suggest changing ‘Conclusion’ to ‘Discussion’. Much of the author's discussion is speculative, but there is still a lack of references to back it up, and the logic of some of the discussion is still lacking.

We appreciate the reviewer's comment about the need for stronger substantiation in our discussion. We have renamed the Conclusions section as “Discussion” and we have thoroughly revised this section to address this concern by adding 20 new references that connect our findings to established literature (see new Bibliography). These additions provide context for the regional variations we observed in FHN patterns and strengthen the logical flow of our arguments.

Specifically, we have:

1. Added references that document global forest fragmentation and urbanization trends to contextualize our observed FHN patterns
2. Connected our findings on European forest transition dynamics to established literature on reforestation efforts and demographic shifts
3. Supported our observations about Southeast Asia, Middle Africa, and South America with references to documented land-use changes in these regions
4. Provided citations for the socioeconomic drivers behind the distributional shifts in FHN we identified
5. Added references on the implications of changing human-forest proximity for ecosystem services and human well-being

These enhancements ensure that our discussion is firmly grounded in existing research while clarifying the logical connections between our observations and their interpretations. We believe these changes substantially strengthen the manuscript and address the reviewer's concern about speculative discussion.

13. I think the discussion should further discuss the linkage between the three indicators ‘FAP’, ‘FPP’, and ‘FHN’ chosen in the manuscript, and how the three indicators can be linked to serve the global ‘FAP’, ‘FHN’, and ‘FHN’. How the three indicators can be linked to serve the interpretation of the spatial proximity of ‘Human-Forest’ globally. I look forward to the author's discussion on how the three subsystems are applicable in different regions of the world.

We thank the reviewer for this insightful suggestion to further discuss the linkages between our three indicators. We have added a new paragraph to the Discussion section (before the final paragraph) that explicitly addresses the relationships between FAP, FPP, and FHN, and how they complement each other to provide a more comprehensive understanding of human-forest spatial relationships.

In this new addition, we discuss:

1. How each indicator captures different aspects of the human-forest relationship
2. The varying regional applicability of each metric, with examples of where each indicator may be most informative
3. How analyzing these indicators together reveals patterns that would not be apparent from any single metric
4. Regional examples demonstrating how the combined interpretation of these indicators provides insights into the spatial dynamics of human-forest interactions
5. The implications for policy development and sustainable landscape management

This addition strengthens the manuscript by clarifying the conceptual integration of our metrics and demonstrating their complementary nature in assessing human-forest proximity across different global contexts. The paragraph is supported by three additional references that address landscape integration approaches and forest spatial dynamics.

14. I suggest adding the limitations and perspectives of the article after Line 254. The key point is the applicability of the new definition (FHN) developed by the authors, its extensibility, and how it can be integrated into the work of ‘Forest-Human’ two-way complex interactions, such as ‘Socio-Economic-Environmental’ synergistic SDGs (Sustainable Development Goals), rather than being used only in the context of the FHN. The potential contribution of this research can be enhanced by integrating it into the work of ‘Forest-Human’ bidirectional complex interactions, such as ‘Socio-Economic-Environmental’ synergistic Sustainable Development Goals (SDGs), rather than just assessing the spatial proximity of people to forests.

We thank the reviewer for this valuable suggestion to expand our discussion on the limitations and broader applicability of the Forest Human Nexus (FHN) metric. Following this recommendation, we have added a substantial new paragraph after the sentence "These patterns may emphasize a need for targeted, regionally specific policies that promote sustainable land-use practices and re-establish connections between human populations and forest ecosystems."

In this new addition, we first acknowledge key limitations of our approach, including constraints in forest definition and the fact that proximity alone doesn't capture actual forest resource use or dependency. We then elaborate on the extensive potential applications of the FHN framework beyond spatial analysis, particularly emphasizing how it can be integrated with socioeconomic and environmental data to address complex bidirectional forest-human interactions.

We explicitly discuss how the FHN can contribute to monitoring progress toward multiple Sustainable Development Goals (SDGs 1, 2, 3, 6, 13, and 15) and provide specific examples of how integrating FHN with other datasets (poverty data, biodiversity information) can inform more equitable and effective policy design. This addition transforms the perspective of our work from a purely spatial analysis to a foundational approach that can inform the complex socio-economic-environmental interactions central to sustainable development (see lines 330-349 of the revised manuscript) .

The revised text is supported by three additional references that address ecosystem services sustainability, biodiversity conservation, and coupled human-natural systems, strengthening the scientific foundation of our discussion.

15. In Lines 255-260, the authors describe the contribution of the article, which in my opinion still needs to be specified, and I believe that the authors should further consider the potential intermediate role variables, i.e., 'Human-X-Forest', which are crucial to the discussion of the human-forest relationship. I suggest that the authors look for appropriate process variables such as forest ecological services, structure and function, integrity and diversity. Emphasise the influence of human activities or forest ecosystems on the mediating variable, which in turn influences another variable.

We appreciate the reviewer's insightful suggestion to address intermediary variables in the human-forest relationship. Following this recommendation, we have substantially revised the paragraph discussing the implications of our findings to explicitly incorporate the "Human-X-Forest" conceptual framework (see new Discussion section).

In the revised text, we identify and discuss three key categories of intermediary variables:

1. Forest structural characteristics (fragmentation patterns, edge density, core habitat area) that mediate between human activities and forest ecological integrity
2. Ecosystem services (carbon sequestration, water regulation, pollination, non-timber forest products) that translate forest proximity into tangible human benefits
3. Governance systems and institutional arrangements that either facilitate or hinder sustainable human-forest interactions

We explain how these mediating variables create mechanistic pathways through which spatial proximity (as measured by FHN) translates into ecological and social outcomes. The revised discussion illustrates how declining FHN values may trigger cascading effects through these intermediary processes, affecting human wellbeing despite physical proximity.

This enhancement provides greater specificity about the contribution of our research while acknowledging the complex, multi-faceted nature of human-forest relationships. We've supported these additions with four new references addressing forest fragmentation, ecosystem services, forest governance, and nature's contributions to people, strengthening the scientific foundation of our discussion.

16. Line 266, the moving window approach still needs to be supported by references or expanded to describe it to ensure experimental reproducibility.

We thank the Reviewer for this comment. We have added relevant references that support our moving window methodology, including a citation. Additionally, we have expanded the method description in the text to provide clearer detail on the window size selection rationale and computational procedure. To ensure full reproducibility, we have made all our analysis code publicly available in the GitHub repository mentioned in the Data Availability section (https://github.com/emanuelemassaro/Forest_Human_Nexus). This repository contains the complete workflow, from data processing to visualization, allowing interested researchers to replicate our analyses or adapt our approach to similar studies.

17. Line 293, different forest definitions can lead to high or underestimation of FPP numbers, I think the authors could, in the method limitations section (Line 313), succinctly state the potential risk that forest definitions can lead to uncertainty in the results.

We thank the reviewer for highlighting this important source of uncertainty. We have added a new paragraph to the Methods limitations section that explicitly addresses how different forest definitions can affect our results.

The added text discusses how the HILDA+ dataset impacts our analysis (see lines 407-445 of the revised manuscript), noting that alternative definitions with different structural thresholds could lead to either over- or underestimation of FPP numbers and spatial patterns. We also acknowledge that the inability to distinguish between natural forests and plantations represents another source of uncertainty in interpreting human-forest relationships.

We've supported this discussion with three relevant references that address forest definition variability, its impact on forest area measurements, and the challenges of translating forest definitions into remote sensing analysis. This addition provides important context for interpreting our results and comparing them with other studies.

Reviewer #2 (Remarks to the Author):

This is a really interesting article building upon the previous research on FPP.

The text is well written and organized, I am just not really happy about the figures and the general conclusions.

We sincerely thank you for your positive assessment of our article. We appreciate your recognition of how our work builds upon previous research on Forest Proximate People (FPP). Your comment about the quality of the writing and organization is encouraging.

We have taken note of your concerns regarding the figures and general conclusions. Based on your feedback and that of other reviewers, we have substantially revised our conclusions section to provide more context, stronger connections to broader implications, and better integration of our three indicators. We have also worked to enhance the figures to improve clarity and information presentation.

We value your input and believe these revisions have strengthened the manuscript considerably. Thank you again for your constructive feedback.

- First, the figures were far more detailed and clearer in the publication from 2020 - to really discern changes or variations in the maps.

We thank the reviewer for this comment. We have increased the resolution of the Figures.

- There are some really interesting conclusions, but feel it could have more impact and relevance when contrasted with auxilliary information such as, where are the expected impacts of climate change? what are the variations in regions in terms of dependence on agriculture for food security? What are the effects of development indices or economic development level?

We appreciate the reviewer's suggestion to increase the impact of our conclusions by incorporating auxiliary information that provides broader context. Following this recommendation, we have substantially expanded our Conclusions section to address these dimensions.

Specifically, we have added (see revised Discussion section):

1. A new paragraph discussing the linkages between our three indicators (FAP, FPP, and FHN) and their complementary roles in understanding human-forest relationships in different global regions. This addition explains how these metrics can be integrated to provide more nuanced insights for region-specific landscape management approaches.
2. A detailed discussion of limitations and future applications of the FHN metric, particularly how it can be integrated with socioeconomic data to monitor progress toward multiple Sustainable Development Goals (SDGs 1, 2, 3, 6, 13, and 15). We explicitly address how our framework can be expanded beyond spatial proximity analysis to contribute to understanding complex social-ecological systems.
3. A significant enhancement addressing the role of intermediary variables in human-forest relationships. We now discuss how forest structural characteristics, ecosystem services, and governance systems function as critical mediating factors that translate spatial proximity into ecological and social outcomes.

These additions directly address the reviewer's suggestions by:

- Connecting our spatial findings to broader development contexts and economic factors
- Discussing how our metrics can integrate with climate change vulnerability assessments
- Acknowledging regional variations in forest dependency and resource needs
- Providing a framework for incorporating development indices in future analyses

We believe these enhancements significantly strengthen the manuscript's relevance and impact while maintaining focus on our core contribution of spatial human-forest nexus analysis.

- I am not sure I am really interested in a comparison between Europe and Asia, I would find it far more useful to discuss what it means for future initiatives, for conservation, for the potential of climate change mitigation, conservation initiatives?

Thank you for this insightful comment. We agree that the manuscript would benefit from a stronger focus on broader implications rather than specific regional comparisons. In the revised version, we have removed Figure 4 and relocated the shift analysis to the Supplementary Information (now Figure S19 in SI), as we recognized it did not provide sufficiently relevant information to the main narrative.

Our revised analysis and discussion now address all macro-regions more comprehensively, with enhanced focus on the implications of our findings for conservation initiatives, climate change mitigation potential, and sustainable landscape management. This approach provides a more balanced global perspective while highlighting the practical applications of our research for addressing critical environmental challenges.

- The dataset is really interesting! I feel it could be assessed in much more interesting ways...and some of the conclusions could be validated with literature or case studies.

We greatly appreciate your enthusiasm about our dataset and its potential. Thank you for recognizing the value of the data we've compiled for analyzing the Forest-Human Nexus.

We agree that there are many additional interesting analyses that could be performed with this dataset. Following your suggestion, we have substantially expanded our conclusions section to better connect our findings with existing literature and have added numerous references to validate our observations. We have incorporated case studies from different regions (particularly Southern Europe and Southeast Asia) to illustrate contrasting trends in forest-human relationships.

Your comment inspires us to consider future work that could explore additional dimensions of these data. The publicly available repository we've created (https://github.com/emanuelemassaro/Forest_Human_Nexus) will enable other researchers to build upon our work and perform complementary analyses that may reveal further insights into human-forest interactions.

Thank you for this constructive feedback that has helped strengthen our manuscript.

Minor comments are in the pdf.

In the revised manuscript, we have addressed all the comments, including the minor comments noted in the PDF. We carefully reviewed each suggestion and made appropriate adjustments throughout the text to improve clarity, accuracy, and presentation of our research.

Overall I would accept this but with some more insights in the discussion and conclusions.

We thank the reviewer for their recommendation and supportive assessment. We agree that enhancing the discussion and conclusions sections would strengthen the manuscript. Based on this feedback and other reviewer comments, we have substantially expanded these sections to provide deeper insights into our findings.

Specifically, we have:

1. Added a comprehensive discussion of how our three indicators (FAP, FPP, and FHN) complement each other and offer different advantages in various regional contexts (see revised Results section).
2. Included a paragraph addressing limitations of our approach and outlining how the FHN framework can be integrated with socioeconomic data to contribute to Sustainable Development Goals (see lines 330-349 of the revised manuscript)
3. Enhanced our analysis of regional variations with additional references to validate our observations (see new Bibliography)
4. Incorporated more discussion of the implications of changing human-forest relationships for ecosystem services and human wellbeing (see revised Discussion section)
5. Added a forward-looking perspective on future research directions (see revised Discussion section)

These additions provide a more nuanced interpretation of our findings and place them in broader context, addressing the reviewer's suggestion for more insights in the discussion and conclusions.